Comprehension Exercise
Electricity

Read the following passage of text and answer the questions.

1 If you examine a television screen you may find it has a thin layer of dust on it but there is no dust on the surrounding furniture. Why is there dust on the TV screen but no dust on the furniture ?

2 Why is a metal conductor connected between the fuel tanker and the plane ?

3 When spraying paint the paint particles are given a positive charge. Why doesn't the wall being painted need to have a negative charge on it for the paint to be attracted electrostatically ?

4 Why is there a discharge between the door handle and the person despite the fact the door handle is not charged ?

5 Draw a circuit diagram of the electric fence. Use the symbol **C** to represent the cow.

6 If you passed an electric current through water, what two elements would be produced ?

7 Why is it inadvisable to carry petrol in plastic containers in the boot of a car ?

8 If the formula for calculating the flow of charge in a circuit is:

$Q = I \times t$ where Q is charge
I is current
t is time

how much charge flows in a cow if it touches an electrified fence and receives an electric current of 20 mA for 0.1 s ?

Comprehension Exercise

Working With Electricity

Answers

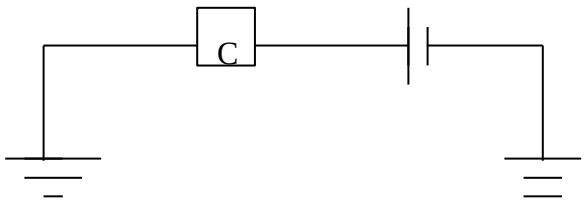
1 The screen becomes charged due to the bombardment of electrons on it over a period of time. The dust particles which may be negatively charged or neutral are attracted to the screen.

2 If there is a build up of charge on the tanker it will be conducted to the earth and not build up to the stage where it will cause a spark and hence an explosion. A build up of charge could be caused by the fuel running through the pipe.

3 Negative charges will be attracted to neutral surfaces as well as positive surfaces.

4 Charge will "jump" to either a neutral point or a point of opposite charge. The door handle does not have to be charged as long as the person is carrying a charge.

5



6 The water would be chemically separated into hydrogen and oxygen. The process is called electrolysis.

7 The plastic container (being and insulator) may build up an electric charge on its surface. If there is a discharge spark the fuel may explode. The charge can be built up by the petrol moving within the container or the outside of the container rubbing against something.

8 $Q = I \times t$

$Q = (20 \times 10^{-3}) \times 0.1$

$= 20 \times 10^{-4}$ coulombs